

Trainings ▾

General Information

The selection of engine coolant **must** be based on the engine model and application, as recommended in this document. Improper selection of coolant could lead to the following:

- Degradation of elastomeric materials
- Corrosion
- Cylinder liner cavitation
- Interaction with cooling system metals.

Coolant types should **not** be mixed in the engine. Mixing coolant formulations could result in critical additive depletion and unpredictable performance. In the case of switching from one coolant formulation to another, drain the cooling system completely and refill with the desired coolant.

Contact the coolant manufacturer for specific guidance if coolants must be mixed.

For most applications, the original equipment manufacturer (OEM) originally filled the cooling system. Check with the OEM on the type of coolant used during the initial fill of the cooling system.

Requirements

Identify a coolant that meets the correct Cummins® engineering standard (CES) for your engine. A list of coolants that meet the appropriate CES for your engine can be obtained from your local Cummins® distributor or can be found on QuickServe® Online (QSOL) at:

Note :

https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html
(https://quickserve.cummins.com/qs3/qsol/service/serviceproducts/coolant_registration.html)

Customers may use either a premixed, heavy-duty coolant or a mixture of high-quality water and heavy-duty concentrated coolant as long as the coolant is approved to the appropriate CES. Refer to Procedure 379-005 in Section 5, (</qs3/pubsys2/xml/en/procedures/00/00-379-005.html>) if **not** using premixed coolant.

Cummins Inc. requires a glycol/antifreeze concentration of 40 to 60 percent by volume for most applications. Cummins Inc. recommends targeting a glycol/antifreeze concentration of 50 percent by volume.

Exceptions:

- In arctic conditions (-32° to -54°C [-25 to - 65°F]) as defined in Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (</qs3/pubsys2/xml/en/bulletin/3379009.html>), and Operation of Automotive Natural Gas Engines in Cold Climate, Bulletin 4332709 (</qs3/pubsys2/xml/en/bulletin/4332709.html>). Cummins Inc. recommends use of a 60 percent

ethylene glycol antifreeze. **Never** exceed 68 percent ethylene glycol concentration in arctic conditions. Do **not** use propylene glycol or glycerin coolant in arctic climates.

- Marine engines **must** use a minimum of 25 percent antifreeze/coolant for both initial fill and topping off. For glycol concentrations less than 40 percent, additive concentration must be increased. If additive concentration can not be increased in the case of organic acid technology (OAT) coolants, coolant **must not** be used below 40 percent glycol concentration. Contact the coolant manufacturer to determine appropriate treatment rates and maintenance guidelines.

Any coolant mixture containing greater than 60 percent water by volume is considered "treated water". **Never** use treated water in place of antifreeze/coolant, unless specifically approved, as engine damage can result. Use of treated water can:

- Reduce the level of engine protection against boil over, cylinder liner pitting, water pump cavitation, corrosion, scale and deposit formation, heater core freeze up, and microbial deterioration.
- Decrease engine and vehicle cooling system component life.
- Damage cooled exhaust gas recirculation (EGR) systems.

Topping off the cooling system with an appropriate pre-diluted or 50/50 coolant is critical for maintaining additive concentrations and freeze protection. **Never** top off the cooling system with water **only**. This can lead to deposit formation and inadequate corrosion and cylinder liner protection.

Top off the cooling system using the same coolant formulation used to fill the cooling system.

Fleetguard® coolants are compatible with other coolant formulations with similar additive packages, but mixing coolant formulations such that one or more coolant(s) is present in a concentration greater than 25 percent by volume relative to the "original" coolant could result in critical additive depletion and unpredictable performance. In the case of switching from one coolant formulation to another, drain the cooling system completely and refill with the desired coolant.

Use the following procedure in the Fluids for Cummins® Products Service Manual, Bulletin 5411406, for information on maintenance of coolant and the cooling system. Refer to Procedure 379-006 in Section 5. (</qs3/pubsys2/xml/en/procedures/00/00-379-006.html>)

Table 1 summarizes Cummins® Engineering Standards for Coolant.

Table 1 - Cummins® Engineering Standards (CES) for Coolant	
Relevant CES	Details
CES 14603	Applies to engines with or without silicone elastomers in cooling system plumbing, and with or without copper-brass heat exchangers that may use high lead solder for repairs.
CES 14439	Applies to engines without silicone elastomers in cooling system plumbing, and without copper-brass heat exchangers that use high lead solder for repairs.
CES 14636	Applies to light-duty, parent bore engines with an aluminum cylinder head, without silicone elastomers in cooling system plumbing, and without copper-brass heat exchangers that use high lead solder for repairs.

Table 1 - Cummins® Engineering Standards (CES) for Coolant

Relevant CES	Details
CES 14648	Applies to engines with top-stop cylinder liners in high-duty cycle applications with or without silicone elastomers in cooling system plumbing, and with or without copper-brass heat exchangers that may use high lead solder for repairs.

Table 2 shows the appropriate CES for various engine models and applications of the engine.

Table 2 - Cummins® Engineering Standards (CES) for Coolant and Requirements for Cummins Engines

Engine Family	Relevant CES
Diesel ISV5.0 B Series gasoline engines	CES 14636
Diesel 12 L, ISX12N, ISX12 G Diesel 15 L, X15N On-Highway Diesel L Series Diesel M Series Diesel X10 Diesel and Natural Gas B-Series engines except B4.5 and B Gas Plus QSF3.8	CES 14603 CES 14439
Industrial and Marine QSK/K19, QSK23, QST30, QSK/K38, QSK45, QSK/K50, QSK60, QSK78 and V903	CES 14648
All Other Engines (Including Power Generation QSK/K19, QSK23, QST30, QSK/K38, QSK45, QSK/K50, QSK60, QSK78)	CES 14603

CES 14603 prohibits the use of the additive 2-ethylhexanoate (2-EHA). Coolants within this specification may or may **not** contain nitrites which can become corrosive to aluminum in certain scenarios. Refer to product packaging or the coolant manufacturer for additional information.

CES 14439 is another coolant performance standard. Performance and test requirements are similar to CES 14603, except that 2-ethylhexanoate (2-EHA) is **not** prohibited in coolants registered to this standard. Coolants registered to 14603 can also be used in place of coolants registered to CES 14439. However, coolants registered to CES 14439 are **not** recommended for use in engines greater than 15 liters in displacement due to potential elastomer compatibility issues with 2-ethylhexanoate which can cause swelling, softening and deformation of silicone gasket, o-ring, hose, and seal materials that can lead to critical coolant leaks.

CES 14636 specifies the use of a nitrite-free coolant in engines with critical parts such as the engine block, cylinder head, or heat exchangers constructed of aluminum. Bare aluminum can catalyze the conversion of nitrite to an alkaline product which can contribute to cooling system corrosion. As a result, coolants containing nitrite should **not** be used in engines with aluminum heads or blocks.

CES 14648 prohibits the use of propylene glycol (PG) based coolants that do **not** include the required nitrite and molybdate additives.

Conventional and hybrid supplemental coolant additive (SCA) coolants using nitrite **must** be re-inhibited frequently with supplemental coolant additive (SCA) to provide cylinder liner pitting protection due to the fact that nitrite depletes significantly over time. Along with the maintenance and compatibility issues that nitrite poses, many OEMs and certain countries prohibit the use of nitrite as part of the coolant formulation due to environmental concerns and regulations. Some nitrite-free OAT coolants have been shown to provide adequate liner pitting protection. There are nitrite free coolants registered to CES 14603, 14439, 14648, and 14636.

Last Modified: 27-May-2026
